EXCEL REVIEW CENTER
CP: 0919 822 5048 | 0917 328 4664

excel online®
my.excelreviewcenter.com.ph

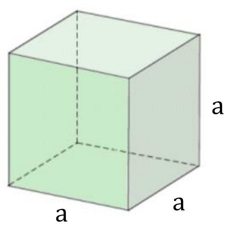
ENGINEERING MATHEMATICS

Solid Geometry 1

Prisms

A prism is a polyhedron comprising an n-sided polygonal base, a second base which is a translated copy of the first, and n other faces joining corresponding sides of the two bases. All cross-sections parallel to the bases are translations of the bases.

1. **Cube** – is a prism with all six faces a square. It is a regular hexahedron and one of the five Platonic solids.



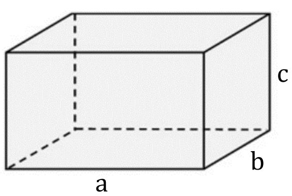
Let V = volume

$$V = a^3$$

Let A_s = surface area

$$A_s = 6a^2$$

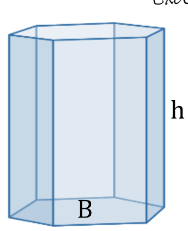
2. **Rectangular parallelepiped** – is a prism with all six faces a rectangle.



$$V = abc$$

$$A_s = 2(ab + bc + ca)$$

3. **Right prims** – is one which has its lateral faces are perpendicular to the base.

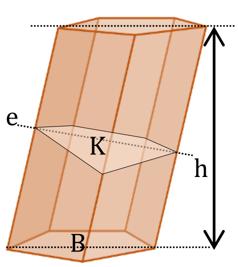


$$V = Bh$$

$$A_L = hp_b$$

Where:
A_L = lateral area
P_b = perimeter of base

4. **Oblique prims** – is one which has its lateral faces are not perpendicular to the base.



$$V = Bh$$

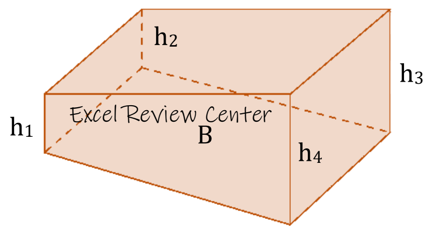
$$V = Ke$$

$$A_L = ep_K$$

Where:
A_L = lateral area
P_K = perimeter of right section
e = lateral edge
K = Area of right section

Note:
Right section is the section perpendicular to the lateral edge of the prism

5. **Truncated prism** – a portion of a prism contained between the base and a plane that is not parallel to the base.

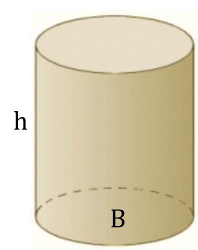


$$V = B\left(\frac{h_1 + h_2 + h_3 + h_4}{4}\right)$$

Cylinders

A cylinder is a solid bounded by a closed cylindrical surface and two parallel planes.

1. **Right cylinder** – is one which has its cylindrical surface perpendicular to the base.

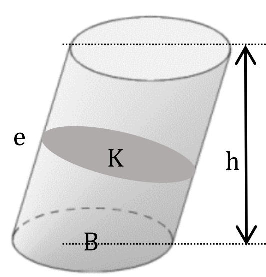


$$V = Bh$$

$$A_L = C_B h$$

Where:
A_L = lateral area
C_B = circumference of the base

2. **Oblique cylinder** – is one which has its cylindrical surface not perpendicular to the base.



Where:
A_L = lateral area
P_K = perimeter of right section
e = lateral edge
K = Area of right section

$$V = Bh$$

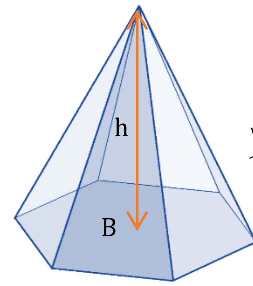
$$V = Ke$$

$$A_L = ep_K$$

Pyramids

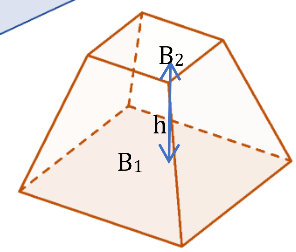
A pyramid is a polyhedron of which one face, called the base, is a polygon of any number of sides and the other faces are triangles which have a common vertex.

A frustum of a pyramid is a portion of the pyramid included between the base and the section parallel to the base.



$$V = \frac{1}{3}Bh$$

Where:
B = area of the base

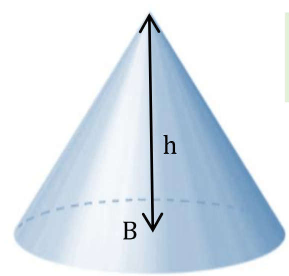


Where:
B_1 and B_2 = area of the parallel bases of the frustum of pyramid

$$V = \frac{h}{3}\left(B_1 + B_3 + \sqrt{B_1 B_2}\right)$$

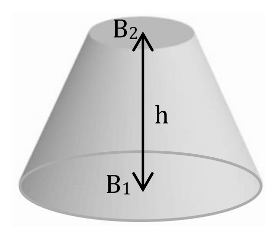
Cone

A cone is a solid bounded by a conical surface (lateral surface) whose directrix is a closed curve,



$$V = \frac{1}{3}Bh$$

Where:
B = area the base



Note:
B_1 and B_2 = area the bases
r = radius of smaller base
R = radius of bigger base

$$V = \frac{h}{3}\left(B_1 + B_3 + \sqrt{B_1 B_2}\right)$$

$$V = \frac{\pi h}{3}\left(R^2 + r^2 + Rr\right)$$

Curved surface area of the frustum of cone, A_s:

$$A_s = \pi(R + r)L$$

Total surface area of the frustum of a cone, A_T:

$$A_T = \pi L(R + r) + \pi r^2 + \pi R^2$$

Where: $L = \sqrt{h^2 + (R - r)^2}$